Thermodynamics of CO_2-loaded aqueous amines with ePC-SAFT. Part II: transfer to MDEA, AMP, piperazine and DEA

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ABSTRACT

The shared water, carbon dioxide and inorganic-ion description established for monoethanolamine in Part I is held fixed while amine-specific parameters are estimated for methyldiethanolamine (MDEA), 2-amino-2-methyl-1-propanol (AMP), piperazine (PZ) and diethanolamine (DEA). The same equation of state accommodates their different reaction networks, but fitted pressures retain loading- and temperature-dependent errors, and PZ species residuals remain large relative to their assumed measurement scale. Held-out partial-pressure average absolute relative deviations (AARD) are 28.19 and 18.27 % for MDEA against Xiao and Ermatchkov, the latter mixing conditions inside and outside the dissociation law's measured temperature range; 42.85 % for AMP against Kundu; 21.94 % for PZ against Ermatchkov; and 79.54 and 44.17 % for DEA against Seo and Hong and Lal.

The data sets and fitting histories differ, so these percentages do not rank the amines. For AMP and DEA, local conditional parameter bands are much smaller than the unfitted-pressure discrepancies; MDEA bands overlap the errors, and PZ sensitivities limit the linear approximation. At least four false liquid splits remain for PZ; a separate fit with Part I's stability penalty still has at least three, two of which the routine phase search missed. DEA reproduces fitted integral absorption heats within 0.98 % AARD but predicts the wrong sign of water–DEA excess enthalpy. The results establish the extent and limitations of amine-specific calibration under a frozen shared description, rather than uniform predictive accuracy across amine classes.

1. Introduction

1.1. From the MEA benchmark to transfer across amine classes

A shared electrolyte description is useful only to the extent that it describes observations beyond those used to estimate it. The companion Part I develops an explicit-ion electrolyte perturbed-chain statistical associating fluid theory (ePC-SAFT) description for loaded aqueous monoethanolamine (MEA), with equilibrium, caloric and density properties calculated from a common thermodynamic potential [1]. Its pressure AARDs at unfitted MEA concentrations are 33.89 % at 15 wt% and 44.53 % at 45 wt%, despite closer agreement at the fitted concentration. Those errors motivate a test of the shared description across other amines; they do not supply a common accuracy standard for different experimental data.

Here the water, CO_2 and inorganic-ion parameters from Part I remain fixed while only amine-specific quantities are fitted. The question is whether amine-specific fitting under this frozen description reproduces observations outside the fit, other thermodynamic properties and single-liquid behavior, and where it does not. MDEA is a tertiary amine whose reaction network contains bicarbonate formation without amine carbamate. AMP is a sterically hindered primary amine, DEA a secondary amine, and PZ a cyclic diamine with mono- and dicarbamate chemistry. These choices test the common bicarbonate description both without carbamate and alongside different carbamate pathways. Part I's bicarbonate–water interaction lies on its upper bound [1], so an amine that forms only bicarbonate tests that shared term most directly. One example per class cannot isolate chemical class from differences in observation coverage, species representation or fitting freedom.

1.2. Calibration, unfitted observations and phase behavior

Published SAFT amine models distinguish parameter estimation from predictions to different extents. Baygi and Pahlavanzadeh fit pure and binary MEA properties and then calculate ternary solubility using ideal-activity chemical speciation [2, Sects. 3.1–3.3]. Nasrifar and Tafazzol also organize MEA, DEA and MDEA around pure

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properties, binary interactions and subsequent acid-gas comparisons, using an ideal-activity treatment rather than explicit electrolyte activity coefficients [3, Sects. 2–6]. Wangler et al. estimate MDEA ion–water interactions using osmotic data before predicting reactive ternary equilibria [4, Sects. 6.1–6.2.1]. The present fits instead include amine-specific reactive observations, so agreement with those observations is calibration.

In this paper, held-out observations never enter a fitting objective. That term does not imply that the data remained unseen throughout selection of the final fit. Previously accessed observations outside the fit and comparisons made after selecting the parameters are identified separately. Finite searches for additional phases provide numerical evidence about the calculated liquids, while experimental pressure and property comparisons measure agreement with the named observations. Neither form of evidence alone establishes independent validation of ion parameters or qualification for absorber design. The results follow the properties being tested, so fitted pressures, unfitted pressures, other thermodynamic properties and phase behavior each constrain the same transfer hypothesis.

2. Thermodynamic model and transferred parameters

2.1. Shared terms and amine-specific chemistry

The residual Helmholtz energy retains the hard-chain, dispersion, association, Debye–Hückel and Born contributions of Part I [1, Sect. 2]. The water and CO_2 component descriptions, hydronium, hydroxide, bicarbonate and carbonate, all interactions among these six shared species, and the reaction laws of Part I, R1 (water autoprotolysis), R2 ($CO_2 + 2\,H_2O \rightleftharpoons HCO_3^- + H_3O^+$) and R3 ($HCO_3^- + H_2O \rightleftharpoons CO_3^{2-} + H_3O^+$), including Part I's fitted R2 corrections, are held fixed. Table 1 distinguishes independent fits, literature values, parameters estimated jointly with MEA, and assumptions. Free interactions between an amine-specific ion and a shared species remain amine-specific parameters; freezing bicarbonate does not freeze every interaction involving bicarbonate.

The MDEA and AMP fits were first developed against an earlier MEA fit whose bicarbonate–water interaction was 0.727 rather than Part I's bound value +1 and whose R2 coefficients differed slightly (Supplementary Section S7); Section 4.4 tests that difference.

Table 1
Transferred quantities and their origins. All shared quantities remain fixed here; amine-specific interactions with shared species may be fitted.

Quantity	Origin in Part I	Implication here
Water segment, energy and association coefficients	Independent fit to IAPWS-95	Fixed; not fitted to any of these amines
CO_2 segment and energy coefficients	Gross and Sadowski, Table 2 [5]	Literature inputs; fixed
CO_2–water dispersion correction and slope	Independent fit to Kiepe pressures	Fixed
Inorganic-ion segment sizes and energies	Uyan et al., Table 4 [6]	Fixed; hydrogen-ion values used for hydronium
Inorganic Born sizes and remaining water–ion terms	Assumed inputs in Part I	Fixed; not independently measured in MEA
Bicarbonate–water interaction	MEA joint fit, upper bound +1	Fixed at bound; conditional SE excludes this direction
R2 level and enthalpy correction	MEA joint fit; interior with conditional errors	Fixed; source-law correction, not a new measurement
R1 and R3 water/carbonate laws	Literature laws as mapped in Part I	Fixed with source-reference conversion
Solvation and dielectric descriptions	Literature coefficients and assumptions in Part I	Water/CO2 inputs fixed; amine coefficients specified separately
Reactive coordinates (MDEA / AMP / PZ / DEA)	Amine-specific fits: 9 / 9 / 12 / 4	Not totals of earlier pure and binary fitting

Each liquid contains the shared species and its molecular amine and protonated forms. AMP and DEA also contain carbamate; PZ contains singly and doubly protonated PZ, mono- and dicarbamate, and the neutral zwitterion HPZCOO. The final reactive fits have nine free coordinates for MDEA, nine for AMP, twelve for PZ and four for DEA,

excluding earlier pure-component, binary and dielectric fits. Supplementary Section S1 gives species parameters, site descriptions, diameter ties and temperature laws. The fitted ion diameters and reaction-law corrections are effective model parameters, not independently measured molecular sizes or equilibrium constants.

Charged species remain liquid-only. The vapor contains water, CO_2 and molecular amine, and for PZ also HPZCOO. Allowing this neutral zwitterion in the vapor resolves otherwise undecided phase searches; with the PZ parameters held fixed, the relative change in fitted partial pressures is at most 1.2×10^{-8}. This numerical change is negligible for those pressures, but the vapor composition space matters to the phase search. PZ has no solid-phase description, and its pure-liquid density is unvalidated.

2.2. Reaction conventions and measured temperature ranges

A state can lie inside a reaction law's declared calculation interval yet outside the temperatures at which that law was measured; Table 2 separates the two. The pure-liquid solvent and infinite-dilution solute conventions follow Part I. The amine laws and shared R2 retain their source molality standard state; R1 and R3 retain the source mole-fraction basis. Explicit water activity and the same hydronium reference are used, with the conversions described in Part I. For a reaction written with stoichiometric coefficients ν_i positive for products, equilibrium requires

$$\sum_i \nu_i \mu_i = 0, \qquad \ln K(T) = a + b/T + c \ln(T/T_0) + dT + \cdots . \tag{1}$$

Source constants are mapped to this reference once; mole-fraction and molality constants are not interchanged without their standard-state conversion. Declared calculation intervals and measured source intervals answer different questions, as shown in Table 2.

Table 2
Reaction-law measurement intervals and use. A calculation inside a declared interval can still extrapolate the measurements. Rounded source limits are distinguished from actual tabulated temperatures.

Law	Measured interval / K	Declared interval / K	Use and limitation
MDEA dissociation	278–368	278–395	393.15 K pressure and 388.71 K heat extrapolate measurements
AMP dissociation	to 353.22	to 353.22	Tong 373–393 K rows not calculated (outside declared interval)
AMP hydrolysis	source slope: 313–353	includes 303–353	Slope extrapolated below 313 K; level and slope fitted
PZ protonation	approximately 293–353	includes 283.15–393.15	283.15 and 393.15 K extrapolate
PZ carbamate	283.15–333.15 (rounded 283–333)	includes 353.15 and 393.15	353.15 and 393.15 K extrapolate
DEA protonation	to approximately 323	to 393.15	23/30 fitted Lawson rows above measured range
Shared R2	not established here	293.15–393.15	Seven Lawson rows at 394.26 K not calculated (outside declared interval); PZ 283.15 K extrapolates below lower limit

The MDEA dissociation coefficients are those of Kamps and Maurer [7], measured over 278–368 K, although the declared calculation interval extends to 395 K. The fixed AMP dissociation law [8] ends at 353.22 K. Its carbamate-hydrolysis temperature dependence is fitted: the fitted b is −1944.8 K, compared with the starting source slope −2546.2 K, while the enthalpy correction reaches its −5000 J/mol limit. This fitted hydrolysis slope is distinct from the fixed dissociation law and does not extend that law's admissible interval. PZ carbamate source measurements span 283.15–333.15 K, and its protonation measurements approximately 293–353 K. DEA uses the fixed Bower protonation law [9] above the approximately 323 K upper measurement limit, including 23 of its 30 fitted pressures. A fitted offset does not turn an extrapolated temperature dependence into a measured one.

3. Data, parameter estimation and numerical assessment

3.1. Observation roles and composition basis

Loading is $\alpha = n_{CO_2,added}/n_{amine,total}$, per mole of amine molecules, including for PZ. Unloaded mass fraction is $w_A = m_A/(m_A + m_w)$; solute-free stock compositions are retained for pressure calculations. True liquid mole fractions use the total number of molecular and ionic species, whereas PZ nuclear magnetic resonance (NMR) group fractions use total PZ as denominator. The protonated-carbamate group and dicarbamate group are therefore not directly comparable to DEA true mole fractions. Total pressure and CO_2 partial pressure are separate observables.

Table 3
Observation roles and prior access. Held out means never in any fitting objective. Comparison-only means calculated after parameter selection without entering that selection. Prior scoring is stated even when no held-out value determined the objective.

Amine / role	Observations	Scored before final fit chosen?
MDEA fitted	278 entries: 67 Kuranov; 16 Jakobsen; 40 Carson; 4 Merkley; 33 heat capacities; 33 loaded and 24 unloaded densities; 61 Kim binary pressures	—
MDEA held-out	20 Xiao and 88 Ermatchkov partial pressures	Yes; previously accessed in model development
MDEA exclusions	Kuranov source rows 28 and 72; seven states above the 393.15 K R2 limit	Not counted as available predictions
AMP fitted	156 entries: 47 Hartono loaded total pressures; 52 Hartono binary entries; 57 Ciftja fractions	—
AMP held-out	51 Kundu partial pressures	Yes; scored during the sequence preceding final diameter choice
AMP held-out comparison	17 Tong total pressures scored; 23 not calculated	Earlier fits had been compared; no blind test claimed
PZ fitted	294 entries: 62 Hilliard pressures; 232 NMR group observations at 121 states	—
PZ held-out	52 Ermatchkov partial pressures	Yes; unconstrained refit scored before penalty alternative and final choice
DEA fitted	80 entries: 30 Lawson pressures at 25 wt%; 50 Carson heats	—
DEA held-out	16 Seo and Hong; 44 Lal partial pressures	Yes; first selected start, then numerically tied second start
DEA comparison-only	Lawson Table VIII; Cai binary; Böttinger species; Horstmann excess heat; Weiland densities	No; added after fit selection

MDEA uses 278 objective rows: Kuranov et al. total pressures [10], Jakobsen et al. protonation fractions [11], Carson et al. [12] and Merkley et al. [13] heats, Weiland et al. heat capacities [14] and loaded densities [15], Sobrino et al. unloaded densities [16] and Kim et al. binary pressures [17]. Its Xiao [18] and Ermatchkov [19] pressure sets had been accessed during earlier development and are held-out assessments rather than new blind tests. AMP uses 47 loaded total pressures [20], 52 binary observations [21] and 57 carbamate fractions [22]; its Kundu set [23] was scored before the final diameter restriction was chosen. PZ uses 62 Hilliard partial pressures [24] and 232 NMR group observations from 121 source states [25]. Its Ermatchkov pressure set was scored on the unconstrained refit before the penalty alternative was examined and the reported fit was finalized. DEA uses 30 Lawson partial pressures [26] at 25 wt% and 50 Carson integral heats at 298.15 K; its held-out sets were first scored on one start and then on a numerically tied second start. Thus held-out here has the same objective-exclusion meaning for every amine, with prior access reported explicitly.

Carrier gases are omitted from the bubble calculations: the model compares $y_{CO_2}P$ with measured partial pressure rather than imposing a nitrogen- or methane-containing experimental gas. This affects the MDEA and PZ nitrogen qualifications and the unreported methane pressure in the Lawson DEA cell. Lal's DEA stock, 2.0 kmol/m^3, is represented by mass fraction 0.2058 using unloaded-solution densities. Böttinger DEA comparisons [27] use 2.5 MPa because source cell pressures span 5–25 bar without individual values. PZ NMR solvent masses reported for heavy

water are treated as water, leaving isotope effects unquantified. These assumptions are part of the compared states, not corrections inferred from the pressure errors.

3.2. Objectives, selection and reported deviations

Each bounded least-squares fit minimizes its original weighted sum,

$$C = \frac{1}{2}\boldsymbol{r}^{\mathsf{T}} W \boldsymbol{r}, \qquad r_j = (y_j^c - y_j^o)/s_j \quad \text{or} \quad r_j = \ln(y_j^c/y_j^o)/s_j. \tag{2}$$

Here s_j denotes the assigned observation scale and W includes group weights or correlations. The objectives use different observations and scales, so their costs do not measure relative physical accuracy across amines. For DEA, pressure log ratios use scale 0.09 and heat differences use 0.02 times the magnitude of measured heat, with each group's squared residuals averaged. For PZ, pressure uses an operational 2 % scale and NMR uses correlated area-normalization errors with assumed area scale $q = 0.02$. Supplementary Section S2 specifies the other residuals and their data definitions.

The pressure AARD definition is identical to Part I,

$$\text{AARD} = \frac{100\,\%}{n}\sum_{j=1}^{n}\left|y_j^c/y_j^o - 1\right|, \qquad \text{bias} = \frac{100\,\%}{n}\sum_{j=1}^{n}(y_j^c/y_j^o - 1). \tag{3}$$

The same reporting formula is applied separately to nonzero heats and other properties; species differences and root-mean-square errors retain their observable units. A negative pressure bias means underprediction. For negative absorption heats, the signed difference $y^c - y^o$ instead states directly whether absorption is more or less exothermic.

The AMP results use the fit whose carbamate diameter is bounded below by the MEA carbamate diameter, 5.32 Å; a fit allowed down to 3.5 Å lowers the cost only slightly, so the data identify the diameter only weakly below about 5.3 Å, and that fit is a sensitivity case. The PZ fit stops at its 100-iteration limit with nearly flat cost and is not labeled formally converged. DEA's two charged-chain starts differ in cost below the optimizer tolerance and represent a numerical tie rather than distinct physical optima. The selected MDEA fit was chosen partly for its held-out pressures and gives poorer agreement with Carson's heats than the earlier fit (scaled RMS 1.106 against 0.977). Supplementary Section S3 reports these selections and their parameter identities.

3.3. Phase searches and conditional uncertainty

Reacting charge-neutral tangent-plane searches follow Part I, requiring a negative distance and a material-conserving Gibbs-energy decrease to identify an additional phase [1, Sect. 3]. Unlike the penalized MEA fit of Part I [1, Sect. 4.5], the reported MDEA, AMP, PZ and DEA fits contain no stability penalty; their phases were searched after fitting. A separate PZ fit applies that penalty and is reported only to show its remaining false liquid splits and the finite search's missed solutions. All split counts are lower bounds; absence of a detected split is not proof of stability. The sampled states and unavailable states remain distinct from a continuous phase boundary.

Local covariance holds shared inputs, model choices, residual weights and bound coordinates fixed. For interior coordinates, the covariance is the local inverse weighted information matrix multiplied by the retained residual-variance factor. At a prediction with parameter derivative g, the conditional variance is $g^{\mathsf{T}} C_\theta g$; displayed intervals use ± 1.96 conditional standard errors (SE), not total prediction intervals. Separate sensitivities displace one marginal SE at a time and do not include correlation compensation. Inward profiles fix a bound coordinate 1, 5 or 10 % of its original interval inside the bound and refit the remaining coordinates. Those displacements have different meanings and sizes; none quantifies source systematics or model-form error.

4. Results: extent and limits of transfer

4.1. What amine-specific calibration recovers

Amine-specific fitting recovers parts of the targeted behavior, but the remaining deviations are larger than several of the assigned observation scales (Table 4). MDEA fitted Kuranov total-pressure AARD is 20.44 %, while binary-pressure AARD is 3.23 %. Its source-scaled RMS exceeds unity for protonation fraction, both heat sources, heat capacity and both density groups. For Carson heat, scaled RMS increases from 0.977 with the earlier shared description to 1.106 after refitting to the Part I shared description. This compromise accompanies recovery of held-out pressures rather than uniformly improved properties.

Table 4
Calibration deviations. AARD uses each observable separately. Scaled RMS is the root mean square of residuals divided by their assigned observation scales, before applying additional objective weights. Pressure residuals are logarithmic for MDEA and relative differences for PZ. These statistics do not provide a cross-amine accuracy ranking.

Observable and fitted source	n	AARD / %	Scaled RMS
MDEA: Kuranov total pressure	67	20.44	0.22
MDEA: protonated fraction	16	13.58	1.64
MDEA: Carson heat	40	5.45	1.11
MDEA: Merkley heat	4	8.46	2.02
MDEA: C_P	33	2.59	1.76
MDEA: loaded density	33	0.18	1.59
MDEA: unloaded density	24	0.06	1.44
MDEA: binary pressure	61	3.23	0.04
AMP: Hartono total pressure	47	10.33	–
PZ: Hilliard partial pressure	62	19.16	12.90
DEA: Lawson partial pressure	30	28.63	–
DEA: Carson integral heat	50	0.98	–

AMP fitted total-pressure AARD is 10.33 %, with bias −0.90 %, and carbamate-fraction RMS is 0.00631. The fitted PZ Hilliard partial pressures give 19.16 % AARD and −3.23 % bias, but RMS is 12.9 times the operational pressure scale. Its mean absolute monocarbamate- and dicarbamate-group errors are 0.0258 and 0.0115 per total PZ, and the NMR whitened RMS remains 11.58 under the assumed area-error model. Thus a moderate mean relative pressure error does not mean agreement at source resolution or an adequate species fit. The separately fitted PZ–water binary properties also retain their discrepancies; they are unchanged by the reactive refit.

DEA fitted pressure AARD is 28.63 %, with bias −0.76 %, whereas integral-heat AARD is 0.977 %. Only 10 of 30 Lawson pressures satisfy $|\ln(p^c/p^o)| \leq 2(0.09)$, twice the assigned log-pressure residual scale, and each isotherm's highest-loading pressure is 31–58 % low. The small aggregate bias therefore conceals loading-shape error (Fig. 1). Forty-six of the fifty Carson heats lie within the assigned 2 % scale. Heat agreement is restricted to these 298.15 K measurements and does not resolve the independent mixing-heat discrepancy below.

4.2. Unfitted pressures and reaction-law extrapolation

Unfitted pressure errors change with source and temperature; the largest occur in extrapolated blocks, but for MDEA, AMP and DEA they already reach 28–52 % AARD at conditions inside the measured law ranges. Held-out MDEA partial-pressure AARD is 28.19 % over twenty Xiao observations at 23.5 wt% and 298.15–313.15 K, all inside the dissociation law's measured range. Every Xiao pressure is underpredicted. The Ermatchkov aggregate, 18.27 % over eighty-eight observations at no more than 49 wt%, mixes measured-law and extrapolated conditions. Inside 278–368 K, its sixty-two pressures give 16.31 % AARD and −11.59 % bias; the twenty-six pressures at 393.15 K give 22.94 % and +21.93 %, respectively. Fitted Kuranov total pressures exhibit the same sign reversal: 11.80 % AARD with −7.80 % bias inside the range, and 31.09 % with +31.09 % bias above it (Table 5). This is an observation about the calculated results, not a diagnosis of the dissociation law; no calculation isolates its contribution from the other temperature-dependent terms.

Table 5
Fitted MDEA errors separated against the 278–368 K dissociation measurements. Heat signed errors are calculated minus measured; they are not relative biases.

Observable	Law range / T	n	AARD / %	Bias / signed error
Kuranov total pressure	inside	37	11.80	−7.80
Kuranov total pressure	outside	30	31.09	31.09
Merkley heat (20 and 40 wt%)	inside (333.15 K)	2	10.36	-5725 J/mol
Merkley heat (20 and 40 wt%)	outside (388.71 K)	2	6.56	-3989 J/mol

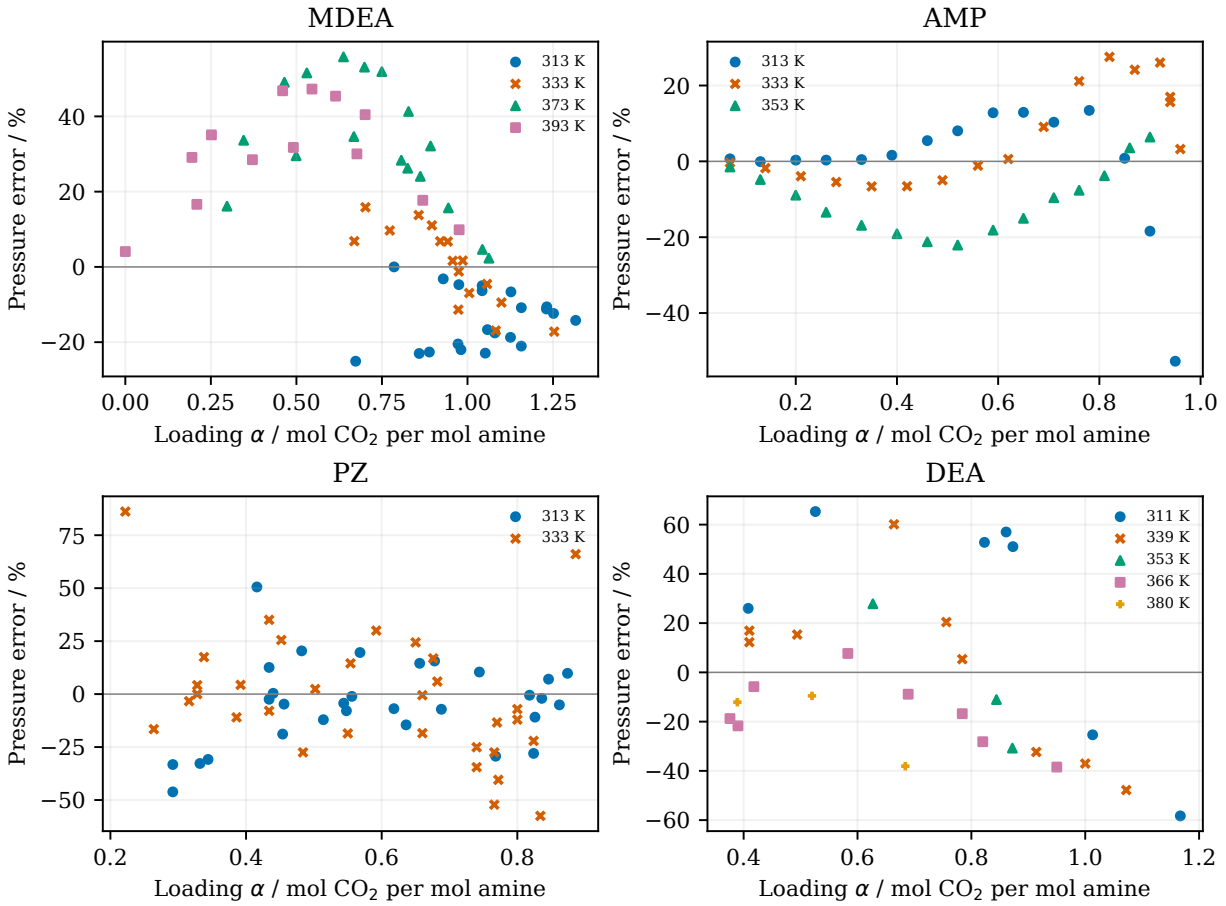


Figure 1: Fitted pressure errors against analytical loading. MDEA and AMP use total pressure; PZ and DEA use CO_2 partial pressure. Each point is a calculation at a measured state. Colors distinguish source temperatures within each panel; no model curve is inferred from the discrete points. MDEA points at 373 and 393 K lie above the 368 K limit of the dissociation measurements.

Table 6
Unfitted pressure errors by source and temperature. MDEA sets were previously accessed; AMP Kundu and PZ Ermatchkov were scored before final parameter selection, and DEA was first scored on a numerically tied fit. All are partial pressures except AMP Tong.

Amine and source	Conditions	*n*	AARD / %	Bias / %	Measured-law range
MDEA Xiao	all	20	28.19	−28.19	inside
MDEA Ermatchkov	all	88	18.27	−1.68	mixed
MDEA Ermatchkov	278–368 K	62	16.31	−11.59	inside
MDEA Ermatchkov	above 368 K	26	22.94	21.93	dissociation extrapolated
AMP Kundu	all	51	42.85	37.79	mixed
AMP Kundu	303.0 K	25	53.94	53.94	hydrolysis slope extrapolated
AMP Kundu	313.0 K	15	43.78	43.78	inside
AMP Kundu	323.0 K	11	16.35	−7.09	inside
AMP Tong (total)	313–353 K	17	28.66	8.11	inside
PZ Ermatchkov	all	52	21.94	7.82	mixed
PZ Ermatchkov	313.1 K	19	9.32	−8.70	inside
PZ Ermatchkov	353.15 K	18	13.12	−8.64	carbamate extrapolated

Amine and source	Conditions	*n*	AARD / %	Bias / %	Measured-law range
PZ Ermatchkov	393.15 K	15	48.52	48.52	carbamate and protonation extrapolated
DEA Seo & Hong	all	16	79.54	77.80	mixed
DEA Seo & Hong	40 °C	5	28.51	22.95	inside
DEA Seo & Hong	60 °C	6	59.82	59.82	protonation extrapolated
DEA Seo & Hong	80 °C	5	154.22	154.22	protonation extrapolated
DEA Lal	all	44	44.17	−22.93	mixed
DEA Lal	40 °C	17	52.25	−48.61	inside
DEA Lal	100 °C	27	39.09	−6.76	protonation extrapolated

The reported AMP fit gives Kundu AARD 42.85 % and bias +37.79 % over all fifty-one observations. The signed error decreases with temperature: +53.94 % at 303 K, +43.78 % at 313 K and −7.09 % at 323 K. The 303 K set extrapolates the source basis of the hydrolysis slope below 313 K; all three temperatures are within the fixed dissociation law's interval. Tong's seventeen total pressures [28] at 313–353 K give 28.66 % AARD and +8.11 % bias, while twenty-three rows at 373–393 K lie above the dissociation law's declared 353.22 K limit and are not calculated. The Tong biases also decrease with temperature, from +48.55 through +5.73 to −23.19 % at approximately 313, 333 and 353 K. The decreasing signed errors and bound hydrolysis enthalpy correction are relevant to temperature dependence, but do not distinguish model error from disagreement between experiments.

PZ Ermatchkov partial pressures [29] give 21.94 % AARD overall, with strong temperature dependence. At 313.1 K, within the carbamate measurements, AARD is 9.32 %; at 353.15 K, above those measurements, it is 13.12 %; at 393.15 K, above both carbamate and protonation measurements, it reaches 48.52 %.

Only eleven of fifty-two pressures lie within the printed source uncertainty, and the uncertainty-scaled RMS is 11.59. The mean error consequently understates the disagreement relative to the source's measurement resolution.

DEA predicts Seo and Hong's thirty-mass-percent solution [30] with 79.54 % AARD and +77.80 % bias; the 80 °C subset reaches 154.22 %. Lal's more dilute stock [31] instead gives 44.17 % AARD and −22.93 % bias. Even the 40 °C subsets, within the Bower measured temperature range, have AARDs 28.51 % against Seo and Hong and 52.25 % against Lal. The higher-temperature subsets extrapolate that law, with the source-specific errors listed in Table 6. Lal's low-pressure observations, which the source reports as less accurate, remain in the AARD (Supplementary Section S4).

The source tables indicate a possible contribution from laboratory disagreement in DEA. At loading about 0.41, Lawson reports 16.0–16.7 kPa at 65.6 °C and 25 wt%, while Seo and Hong report 11.6 kPa at 80 °C and 30 wt%. These are not matched states: temperature, concentration and loading differ. They do not isolate experimental bias or establish which source is correct, and the model retains a loading-shape error within its own fitted source. Across amines, unmatched pressure types, source uncertainties, concentrations, temperatures and selection histories prevent comparison of their error percentages as an accuracy ranking.

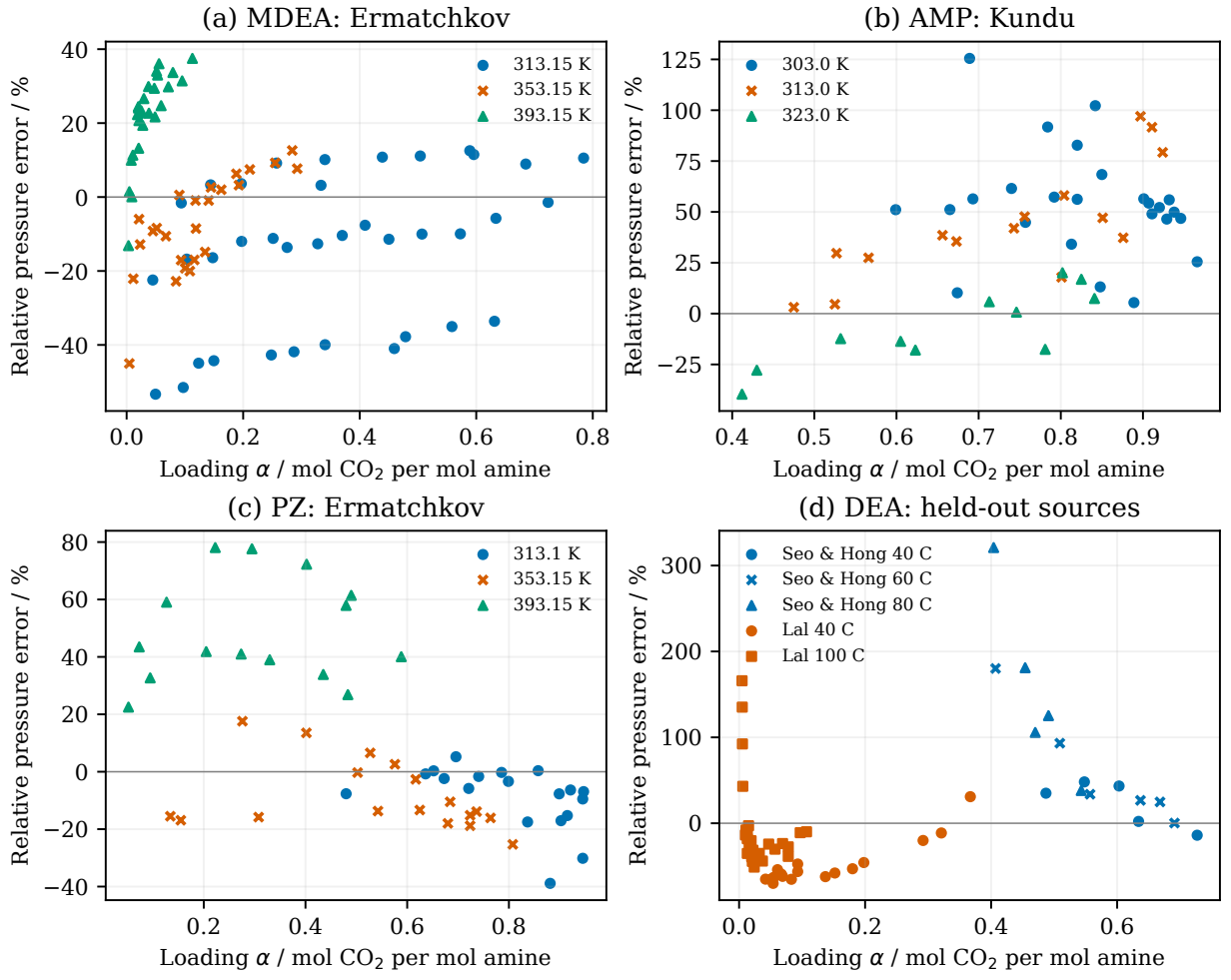


Figure 2: Relative partial-pressure errors at measured loadings. All panels use held-out sets; the MDEA set was previously accessed. MDEA at 393.15 K exceeds its dissociation measurements; PZ at 353.15 K exceeds its carbamate measurements and at 393.15 K also its protonation measurements; AMP at 303 K extrapolates the hydrolysis source slope. DEA marker shapes distinguish 40, 60, 80 and 100 °C. Points from different stocks are shown without interpolating a continuous relation.

4.3. Conditional parameter uncertainty beside the unfitted errors

For AMP and DEA, conditional parameter bands are far smaller than the observed unfitted-pressure errors (Fig. 3). AMP Kundu relative SEs range from 3.2 to 13.5 %, with median 6.7 %; only eight of fifty-one observations fall within the nominal ±1.96 SE intervals. DEA relative pressure SEs are 5.4–11.6 % for Seo and Hong and 12.3–19.1 % for Lal, compared with their much larger measured deviations. These comparisons show that uncertainty within the selected local parameter fit does not encompass the discrepancies. They do not identify which experiment or model term causes a miss.

MDEA bands at the 108 Xiao and Ermatchkov pressure states have relative SEs of 7.82–39.1 %, overlapping the magnitude of the held-out errors. Its cation–bicarbonate interaction and cation Born diameter lie at +1 and 7 Å, respectively, and remain fixed in this covariance. For AMP, the carbamate diameter and hydrolysis enthalpy correction are fixed at their limits, 5.32 Å and −5000 J/mol. The corresponding residual-variance factors are 2.7639784 for MDEA and 0.829 for AMP. The band size therefore depends on both the fitted information and the conditional error model, not just the parameter count.

PZ's local linear uncertainty is less reliable: one marginal SE of a fitted dispersion energy changes some Hilliard calibration pressures by 90–160 % in the linear estimate. Its cation–bicarbonate interaction is −0.02 ± 1.03 across an

allowed interval of width 1.5, indicating weak identification. With the PZH^+–water interaction fixed at its −1 limit, eleven interior coordinates use variance factor 146.3; calibration-pressure relative SEs range from 2.8 to 34 %. No PZ held-out pressure band was calculated, and the calibration band cannot substitute for it. The inward 5 and 10 % profile refits did not converge, so their costs are upper bounds.

DEA has no active parameter bounds, but its hydrolysis-law level and carbamate–water interaction have correlation 0.972. Its residual-variance factor is 0.210, and relative SEs at fitted Lawson and Carson states are 3.3–10.3 % and 0.9–1.9 %, respectively. Small conditional errors coexist with compensation between parameters. Supplementary Section S6 reports all parameter SEs, sensitivity definitions and inward profiles. None includes shared-input uncertainty, experimental systematic errors, uncertainty in the omitted chemistry or model discrepancy.

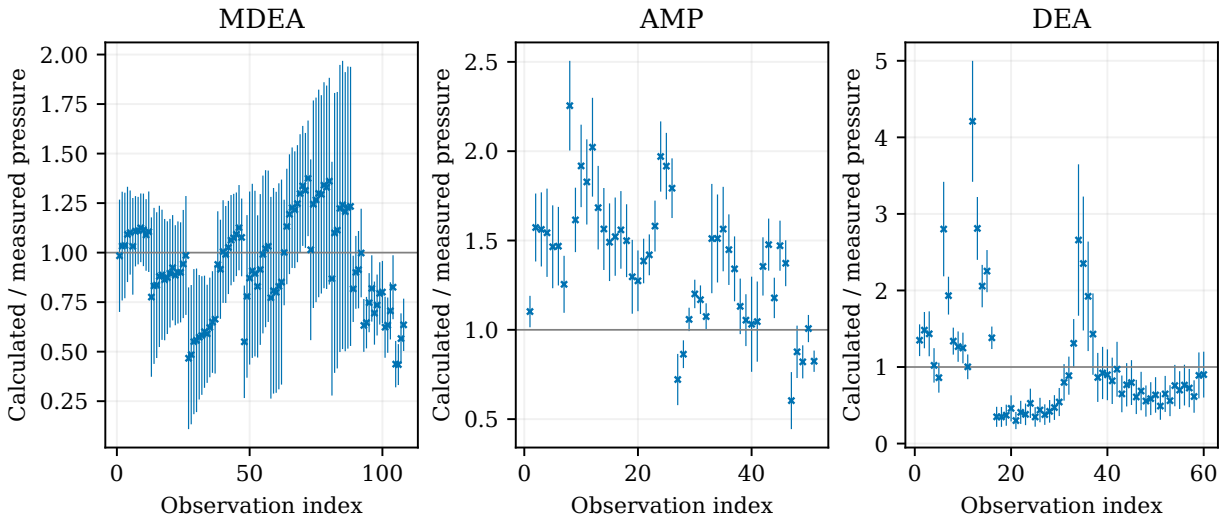


Figure 3: Calculated-to-measured partial-pressure ratios with nominal ±1.96 conditional parameter-SE intervals, at stored measured states. Separate panels show MDEA Xiao/Ermatchkov, AMP Kundu and DEA Seo and Hong/Lal; the horizontal axis numbers the observations within each panel and is not a temperature or loading axis. The horizontal line is equality. Intervals are linear parameter propagation, conditional on fixed shared inputs and active bounds; they exclude measurement noise and model discrepancy. PZ held-out bands are unavailable.

4.4. Sensitivity of unfitted pressures to the shared description

Unfitted pressures depend on the shared description as well as on the amine fit: substituting Part I's bicarbonate–water interaction and R2 law moves the MDEA and AMP errors by 22–26 percentage points in opposite directions before the amine parameters are refitted (Table 7). Table 7 follows the same data through the earlier shared description, substitution of the Part I bicarbonate–water interaction and fitted R2 law, and amine-specific refitting. For AMP, Kundu AARD changes from 43.69 to 21.89 % upon substitution without refitting, then returns to 43.51 % when the original AMP coordinates and limits are refitted. The substitution worsens fitted total-pressure AARD to 18.85 %, while refitting recovers 9.93 %. The favorable 21.89 % score thus reflects cancelling changes in this comparison, not a jointly improved calibration and independent pressure description.

Table 7
Same-data pressure AARD (%). Each row compares the earlier shared inputs, substitution of the Part I bicarbonate–water interaction and R2 law without refitting the amine, then amine-specific refitting. A3 is a counter-case, not the selected MDEA description. AMP here uses the original bounds; its final bounded fit is reported separately.

Amine fit (unfitted source)	Earlier	Substituted	Refitted
MDEA alternative ion description A3 (Ermatchkov)	26.21	52.47	31.44
MDEA reported fit (Ermatchkov)	19.15	42.03	18.27
AMP (Kundu)	43.69	21.89	43.51

For the reported MDEA fit, the same sequence changes Ermatchkov AARD from 19.15 to 42.03 and then 18.27 %. An alternative MDEA ion description, A3, changes from 26.21 to 52.47 and then 31.44 %, so refitting does not recover its earlier error. A3 is a counter-case, not an additional selected model. These same-source interventions demonstrate parameter compensation but do not uniquely assign the discrepancy to bicarbonate, Born solvation or a reaction constant.

The reported AMP fit widens the cation–CO_2 interaction and retains the 5.32 Å carbamate floor, giving 42.85 % Kundu AARD. Lowering the floor to 3.5 Å gives a slightly smaller calibration cost but 43.95 % Kundu AARD; it is not the reported result. Supplementary Section S7 gives the complete sequence. A model-free comparison of the Hartono and Kundu measurements lacks sufficient matched conditions, especially at 303 K, to decide whether Hartono and Kundu disagree or whether the model misses their concentration dependence. Its interpolation and vapor-baseline assumptions do not establish that either source is erroneous.

4.5. Comparison with published models

Table 8
Published-model comparisons with data roles stated. Only the PZ Hilliard row has the same fitted source and row count in both models; objective and chemistry still differ. Tong coverage differs. MDEA and qualitative comparisons use different observation subsets.

Model / system	Source and published deviation	Published role / role here
Yi e-NRTL, PZ [32, Table 3]	Hilliard: 62 points, 18.0% partial-pressure AARD	Fitted / fitted
Yi e-NRTL, PZ [32, Table 3]	Ermatchkov: 52 points, 7.8% partial-pressure AARD	Fitted / held out; previously scored
Yi e-NRTL, AMP [32, Table 4]	Kundu: 51 points, 15.4% partial-pressure AARD	Fitted / held out; previously scored
Yi e-NRTL, AMP [32, Table 4]	Tong: 45 points, 8.0% total-pressure AARD	Fitted / unfitted; 17 points here
Wangler ePC-SAFT, MDEA [4, Table 11]	Own 82-point set: ARD 19.7%; corrected Uyan 34.4%	Ternary predictions / different reactive fit and partial-pressure subsets here
Cleeton ePC-SAFT, MDEA [33, Sect. 3.3.2]	Concentration and low-loading comparisons; no matched score	Ternary predictions after pure/binary estimation
Perdomo SAFT-γ Mie, PZ / DEA [34, Sects. 4.2, 4.3.3]	PZ concentration transfer; Lal pressures and Böttinger species	Selected reactive group estimation; other concentrations/species predicted
Nasrifar PC-SAFT, DEA [3, Sect. 6, Figs. 13–14]	Species and acid-gas curves; no matched score	Predicted after pure/binary estimation; ideal-activity chemistry

Only the Hilliard PZ pressures are fitted both here and in a published model, where the e-NRTL deviation of Yi et al. is similar (18.0 against 19.16 %); every other published comparison differs in which observations were fitted. Yi et al.'s e-NRTL model provides two same-source PZ comparisons: 18.0 % partial-pressure AARD for the same sixty-two Hilliard observations, fitted in both models, and 7.8 % for fifty-two Ermatchkov observations, fitted by Yi but held out here [32, Table 3, p. 2]. For AMP, Yi reports 15.4 % on the fifty-one Kundu partial pressures, fitted there and held out here [32, Table 4, p. 5]. Yi neglects AMP carbamate [32, Sect. 2.1.2, p. 3], whereas the present fit includes carbamate and fits its species fraction. Yi's 8.0 % Tong result is total-pressure AARD on forty-five fitted observations, compared with seventeen unfitted total pressures scored here. That unequal coverage is not a matched-data performance comparison.

Wangler et al. report MDEA pressure ARD 19.7 % over their eighty-two ternary observations and correct Uyan's result to 34.4 % on that set [4, Table 11]. Both are ternary predictions after lower-order estimation, including osmotic data for the ion–water interactions in Wangler's work; neither fits those ternary pressures. The present MDEA fit includes reactive total pressures, and its held-out partial-pressure subsets are different, so those percentages cannot establish relative accuracy. Cleeton et al. likewise predict MDEA ternary pressures from pure and binary inputs and discuss larger concentrated-solution and low-loading discrepancies [33, Sect. 3.3.2, Figs. 7–8]. Their separation of data roles and condition-dependent errors is more informative here than a pooled score from unmatched sets.

Perdomo et al.'s SAFT-γ Mie model represents reactions implicitly through association, with selected PZ concentration data used to estimate group interactions and other concentrations predicted [34, Sect. 4.2, Fig. 4]. Its DEA comparisons include Lal pressures and Böttinger species, with pressure and species predictions after the specified group-interaction estimation rather than an independent explicit-ion parameter test [34, Sect. 4.3.3, Figs. 13–14]. Nasrifar and Tafazzol's DEA acid-gas curves are predictions after pure and binary estimation, with ideal-activity chemistry and no explicit electrolyte activity coefficients [3, Sects. 2–6, Figs. 13–14]. These qualitative comparisons distinguish fitting policies and chemistry; they supply no matched-row ranking against the present DEA fit.

4.6. Species, heats, density and binary properties

DEA shows that close agreement in fitted absorption heats and in comparison-only species fractions does not extend to every property: the water–DEA mixing enthalpy has the wrong sign. At nine Böttinger states without the oxazolidone byproduct, mean absolute true-mole-fraction differences are 0.001749 for DEA plus $DEAH^+$, 0.001301 for carbamate and 0.001274 for the bicarbonate-labeled sum. The latter is compared with model bicarbonate plus carbonate. Sixteen byproduct-containing states remain separate because that chemistry is omitted, and no NMR mole-fraction measurement uncertainty is reported. The pressure assumption and aggregate species definitions condition the comparison (Fig. 4).

At 363.15 K and 1.75 MPa, all fourteen water–DEA excess-enthalpy calculations [35] have the wrong sign: approximately +86 to +758 J/mol, against measured values from −93 to −1707 J/mol. None lies within twice the source scale. This error concerns heat of mixing per mole of water–DEA mixture, whereas Carson's fitted heats concern absorption per mole of added CO_2 at 298.15 K. Agreement with the latter cannot establish a correct mixing enthalpy at another temperature. No corresponding excess-enthalpy comparison is available for MEA, MDEA, AMP or PZ, so the sign error has not been shown to be DEA-specific. Cancellation of mixing contributions during absorption is possible but was not calculated.

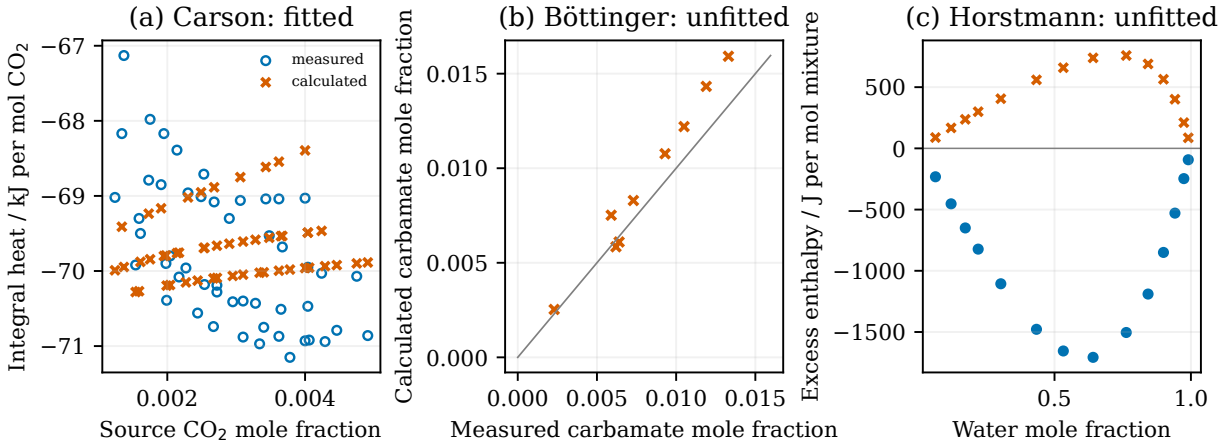


Figure 4: DEA properties at the measured coordinates: (a) fitted Carson integral absorption heats at 298.15 K against the source CO_2 mole fraction, with the source heat-release magnitude converted to negative enthalpy for exothermic absorption; (b) comparison-only carbamate mole fractions in the nine Böttinger states without byproduct, at assumed 2.5 MPa; (c) comparison-only water–DEA excess enthalpy at 363.15 K and 1.75 MPa. Circles are measurements and crosses calculations. The different enthalpy bases and temperatures separate the successful absorption-heat calibration from the wrong-sign mixing prediction.

DEA comparison-only pressure AARD is 41.74 % for Lawson Table VIII, nominally 50 wt% but 54.5 wt% at 77 °F. Cai binary data [36] give 40.82 % pressure AARD at 6.66 kPa, far from the fitted binary isotherm, while forty-three loaded densities at 298 K give 0.445 % AARD. Those properties use different data, conditions and measurement scales (Supplementary Section S4). The MDEA held-out capped absorption heat at 388.71 K and 60 wt% is −64.491 kJ/mol CO_2, against −63.3 kJ/mol; this extrapolates the dissociation measurements and solvent concentration. The corresponding 333.15 K, 60 wt% capped heat could not be calculated. PZ has no comparable reactive heat or density assessment; the four amines have not been checked equally across properties.

4.7. Phase behavior limits the apparent fit

Small property residuals do not exclude false liquid phases: the reported PZ fit has at least four, and a separate fit with the Part I stability penalty still has at least three, two of which the routine search missed. At least four false liquid splits remain for the reported PZ fit at Hilliard rows 55, 56, 61 and 62, near 4.96–5.02 mol PZ/kg water, loading 0.77–0.83 and 313 or 333 K. The calculated additional liquids contradict the measured liquid states despite the moderate pressure AARD. The search examines 183 calibration states, comprising 121 NMR states and sixty-two bubble states, and finds no split at the fifty-two held-out bubble states. Rich NMR states below their calculated bubble pressure are assessed as liquids at 1.05 times that pressure; those moved states do not demonstrate single-liquid behavior at the original assumed 0.1 MPa.

The separate PZ penalty fit does not remove the false liquid splitting. Its routine census detects only one split, but direct tangent-plane evaluations at known split compositions show two further negative distances, $-0.0309\ RT$ and $-0.0437\ RT$, at rows 55 and 56. The penalty fit therefore has at least three false liquid splits. This counterexample demonstrates a missed finite search, so every split count in this paper is a lower bound, and the absence of detected instability, here and for the penalized MEA fit of Part I [1, Sect. 4.5], does not prove stability.

The MDEA searches cover 775 calculated states, some repeating a physical state; none detects an additional liquid, 37 detect a gas-like phase, and none is ambiguous or unresolved. Seven source states above the shared R2 interval are excluded, and there is no phase evidence at 394.94–395.00 K. AMP has no detected liquid, gas, ambiguous or unresolved outcome at the fifty-one Kundu states. Its calibration searches retain no detected liquid, ambiguous or unresolved outcome, but their gas outcomes were not stored. DEA searches the 140 fitted and held-out states without detecting an additional phase or an unresolved result.

Table 9
Finite phase-search results. Detected liquid-split counts are lower bounds.

Fit / states	Detected liquid splits	Other outcomes and omissions
MDEA: 775 searched states	0 found	37 gas-like phases; no ambiguous or unresolved result; seven states above the R2 interval not calculated
AMP: 51 Kundu states	0 found	No gas, ambiguous or unresolved result; calibration gas outcomes not stored
PZ: 183 calibration states	at least 4	No ambiguous or unresolved result; three rich NMR pressures moved above bubble pressure
PZ: 52 held-out bubble states	0 found	Finite search only
PZ penalty alternative	at least 3	One found routinely; two additional splits resolved by direct distance evaluation
DEA: 140 fitted/held-out states	0 found	No gas, ambiguous or unresolved result

5. Discussion: what the common description transfers

5.1. Shared thermodynamics and parameter compensation

The common equation of state, reference conventions and frozen water/carbonate description can be used with different amine reaction networks while fitting only amine-specific quantities. What transfers is that thermodynamic description and a conditional ability to fit selected properties; uniform pressure, caloric and phase accuracy does not follow. MDEA and AMP illustrate how changing the same shared quantities can move unfitted errors in opposite directions before refitting, and how amine-specific compensation can restore an earlier pressure pattern. The alternative MDEA ion description (A3) shows that recovery is not automatic even within MDEA. Substituting Part I's bicarbonate–water interaction and R2 law moves the MDEA and AMP unfitted errors in opposite directions by 22–26 percentage points before refitting, and the fitted MDEA cation–bicarbonate interaction ends on its +1 bound; this bicarbonate-related pattern is an observation across these calculations, not proof of a uniquely defective bicarbonate parameter or Born term.

The different parameter freedoms and observations also matter. PZ needs a more extensive species network, retains a weak cation–bicarbonate interaction and large local energy sensitivities, and predicts false additional liquids. DEA's four reactive parameters have small conditional errors and no active bounds, yet large unfitted-pressure errors remain. AMP's carbamate diameter and enthalpy correction reach limits with nearly flat inward profiles (Supplementary

Section S6). A diameter at a bound is evidence about this constrained fit, not a measurement of an ion size. One solute per chemical class, different data objectives and different phase grids do not distinguish an intrinsic class effect from those other changes.

5.2. Source disagreement, extrapolation and omitted effects

Temperature-range separation narrows the interpretation without assigning cause. MDEA and PZ have larger positive errors in high-temperature blocks that extrapolate measured reaction laws, but MDEA Xiao and DEA's low-temperature held-out subsets already show substantial errors inside those law ranges. AMP's decreasing signed error with temperature occurs while its fitted hydrolysis enthalpy correction is on a limit. These observations indicate conditions where the thermodynamic description is inaccurate; they do not isolate a law slope from an interaction, a solvent-property extrapolation or source disagreement.

The source comparisons also remain incomplete. AMP lacks sufficient matched experimental states to separate the Kundu/Hartono difference from concentration-dependent model error, especially at its coldest isotherm. DEA source values near one loading suggest laboratory disagreement, but those states differ in temperature and concentration and cannot determine which source is correct. Gas dilution, assumed cell pressures, PZ heavy-water treatment and omitted byproduct chemistry have distinct origins and should not be collapsed into one explanation for all pressure errors. The wrong water–DEA mixing-enthalpy sign adds an independent thermodynamic limitation that fitted absorption heats do not erase.

5.3. Scope of the evidence

Within the tested conditions, the fits agree with DEA integral absorption heats at 298.15 K (0.98 % AARD), loaded DEA densities at 298 K (0.445 %) and PZ pressures at 313.1 K (9.32 %). Tests near the PZ false splits (about 5 mol/kg, loading 0.77–0.83), for the water–DEA mixing enthalpy and above reaction-law measurement ranges reveal the limitations reported here; conditions beyond the measured and searched states were not tested. The comparisons support statements about the specified equilibrium and property calculations at their measured and searched states. They combine calibration, fixed-parameter prediction of observations outside the objective, comparison-only properties and numerical phase verification. Previously accessed held-out results do not become blind validation merely because their residuals are absent from the objective. Finite phase searches do not establish a continuous phase boundary, and the PZ counterexample shows that the search may miss a resolved split. The results do not qualify absorber or stripper design, kinetic or transport coefficients, solvent blends, degradation or solid equilibrium. These limits define the demonstrated transfer.

6. Conclusions

A frozen description of water, CO_2 and inorganic ions can accompany amine-specific fitting across MDEA, AMP, PZ and DEA, but calibration and transfer remain property- and source-dependent. Residual loading-shape errors persist in the fitted pressures, and PZ species residuals remain large relative to their assumed measurement scales. Shared-parameter substitutions followed by refitting demonstrate compensation: an apparently improved unfitted AMP score before refitting does not survive restoration of its calibration fit.

MDEA's held-out pressure AARDs are 28.19 % for Xiao and 18.27 % for Ermatchkov, with the latter mixing measured-range and extrapolated dissociation conditions. AMP's held-out Kundu error is 42.85 %, while PZ gives 21.94 % against held-out Ermatchkov pressures and 48.52 % at 393.15 K. DEA gives 79.54 and 44.17 % on the held-out Seo and Hong and Lal data, despite 0.98 % fitted integral-heat AARD, and predicts the wrong sign of water–DEA excess enthalpy.

Local conditional parameter bands are much smaller than the AMP and DEA unfitted discrepancies, overlap the MDEA errors, and are limited by large PZ sensitivities. At least four false liquid splits remain for the reported PZ fit; a separate penalty fit and direct split evaluations demonstrate that the finite phase search can miss instability. No additional phase was detected in the MDEA, AMP and DEA searches. The extent of transfer is therefore defined by the reported calibration errors, unfitted property failures, reaction-law extrapolations and conditional uncertainty, rather than by a universal accuracy or process-use claim.

Data and code availability

Processed observations, parameter files, numerical results and presentation scripts are provided with the public electronic data at `https://github.com/tannerpolley/Amine-Thermodynamics`. Presentation code reads the stored calculations without evaluating the thermodynamic model again. Input checksums and the source of every displayed numerical result are listed in the accompanying numerical-source index. The calculations use the compiled `epcsaft` library identified in Supplementary Section S8. Deposit identifier: *[Zenodo DOI supplied in the final submission]*. Referee access to the compiled ePC-SAFT library: *[access arrangement supplied in the final submission]*. Third-party measurements retain their source terms.

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Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT author statement

Tanner W. Polley: Conceptualization, methodology, software, formal analysis, investigation, data curation, writing–original draft, writing–review and editing, visualization.

Declaration of generative AI and AI-assisted technologies

During preparation of this manuscript, OpenAI Codex and Claude (Anthropic) were used to draft and revise manuscript text, check computational consistency, and assist with source checking. The author reviewed and edited the content and takes responsibility for the final manuscript.

Nomenclature

α: added CO_2 per total amine, mol/mol; w_A: unloaded amine mass fraction; x_i, y_i: liquid and vapor true-species mole fractions; P: total pressure; $p_{CO_2} = y_{CO_2} P$: partial pressure; K: equilibrium constant on the stated standard state; C: least-squares cost; s: assigned observation scale; SE: conditional standard error; AARD: average absolute relative deviation; NMR: nuclear magnetic resonance; k_{ij}: unlike-pair dispersion correction; d^{Born}: Born diameter.

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